IILM UNIVERSITY, GREATER NOIDA

SCHOOL OF COMPUTER SCIENCE & ENGINEERING

PERFORMANCE RESULTS REPORT

Project Title: Predictive Maintenance System for Data Centre Servers

1. Project Details

Project ID	26P_3CSE276-Predictive Maintenance System for Data Centre Servers
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Project Guide	Shubham Jayasval

2. Objective

This report records how well the prediction component of the Predictive Maintenance System for Data Centre Servers performs at flagging servers that need maintenance, using CPU usage, memory usage, temperature and disk usage.

Note: The dataset used is synthetic (generated by the project's script, seed 42), not real data-centre telemetry. All figures below were measured by running the code on that dataset; results on real server data will differ.

3. Model and Evaluation Setup

Models compared	Logistic Regression, Random Forest (200 trees), Gradient Boosting
Selected model	Gradient Boosting (highest F1-score on the test set)
Train / Test split	80% / 20% stratified (4000 train, 1000 test), random_state 42
Cross-validation	5-fold stratified, scored on F1
Target variable	maintenance_needed (1 = maintenance needed, 0 = normal)
Decision threshold	Probability ≥ 0.5 raises a MAINTENANCE ALERT
Libraries / tools	Python 3.13.18, scikit-learn 1.9.1

4. Metrics of Selected Model (Gradient Boosting)

Metric	Value	Remarks
Accuracy	87.70%	Share of all test records classified correctly
Precision	0.841	Of records flagged for maintenance, share that truly needed it

Metric	Value	Remarks
Recall	0.795	Of records that needed maintenance, share that were caught
F1-score	0.817	Balance of precision and recall
ROC-AUC	0.938	Ranking quality across all thresholds
5-fold CV F1 (mean)	0.797	Consistency check across data splits

5. Confusion Matrix (Test Set)

	Predicted: Normal	Predicted: Maintenance
Actual: Normal	602	52
Actual: Maintenance	71	275

Of 346 servers that needed maintenance, 275 were caught and 71 were missed; 52 of 654 normal servers raised a false alarm.

6. Model Comparison (Test Set)

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	87.30%	0.823	0.806	0.815	0.944
Random Forest	85.90%	0.802	0.786	0.794	0.934
Gradient Boosting	87.70%	0.841	0.795	0.817	0.938

7. Feature Importance

temperature 0.530, cpu_usage 0.199, memory_usage 0.151, disk_usage 0.119. Temperature contributes the most to the prediction.

8. Observations and Conclusion

- The three models performed similarly (accuracy 85.90% to 87.70%); Gradient Boosting scored the highest F1 and was selected.
- About 21% of servers needing maintenance were missed at the 0.5 threshold. Lowering the threshold would catch more at the cost of more false alarms.
- The labels in the synthetic data come from a noisy rule, so scores reflect how well the model recovers that rule, not real-world failure prediction. Re-evaluate on real telemetry before relying on these numbers.